

Abacus.AI Platform Mapping Review

Document Purpose: Complete review of Abacus.AI platform UI structure, components, and design patterns extracted from conversation history and PRD documentation.

Date: November 17, 2025

Source: Conversation thread history + PRD files (Real-time Progress Streaming, Multi-LLM Routing, etc.)

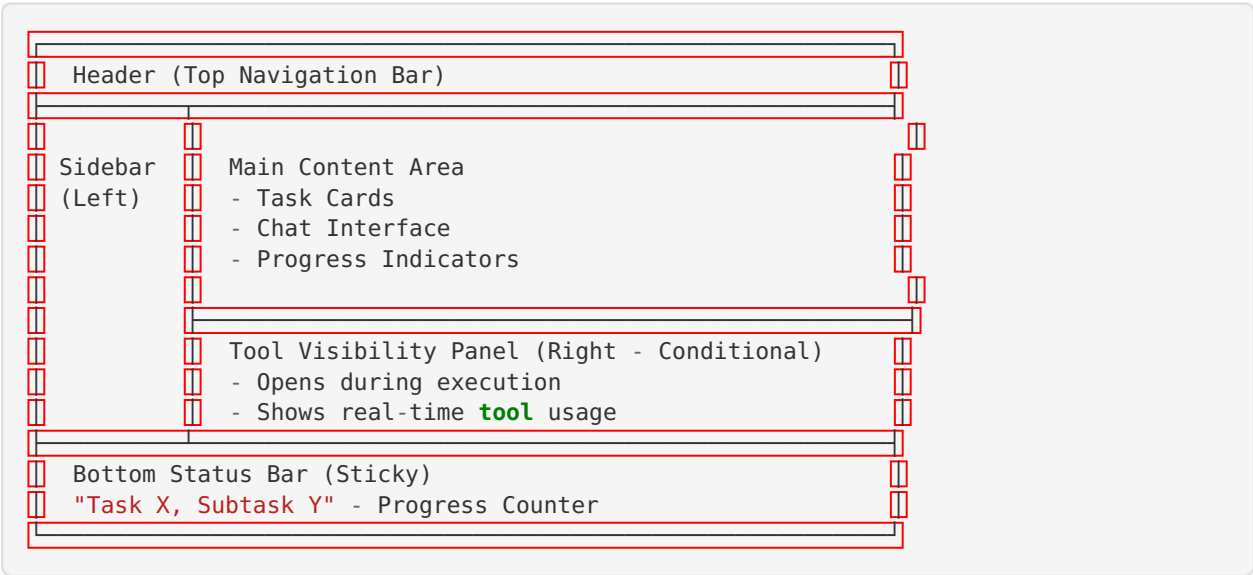
Executive Summary

The Abacus.AI platform (specifically the DeepAgent implementation) follows a **transparency-first design philosophy** where users can see exactly what the AI agent is doing at all times. The platform emphasizes real-time progress visibility, clear status indicators, and comprehensive tool usage tracking.

Core Design Principle: “Transparency Over Abstraction” - show what the agent is doing at all times.

1. Overall Layout Structure

Primary Layout Components



Layout Characteristics

- **Responsive:** Adapts to desktop and mobile viewports
- **Persistent Elements:** Sidebar (left), Status Bar (bottom)
- **Conditional Elements:** Tool Visibility Panel (right) - appears during task execution
- **Main Content:** Scrollable, contains task cards and chat interface

2. Header Component

Structure



Key Elements

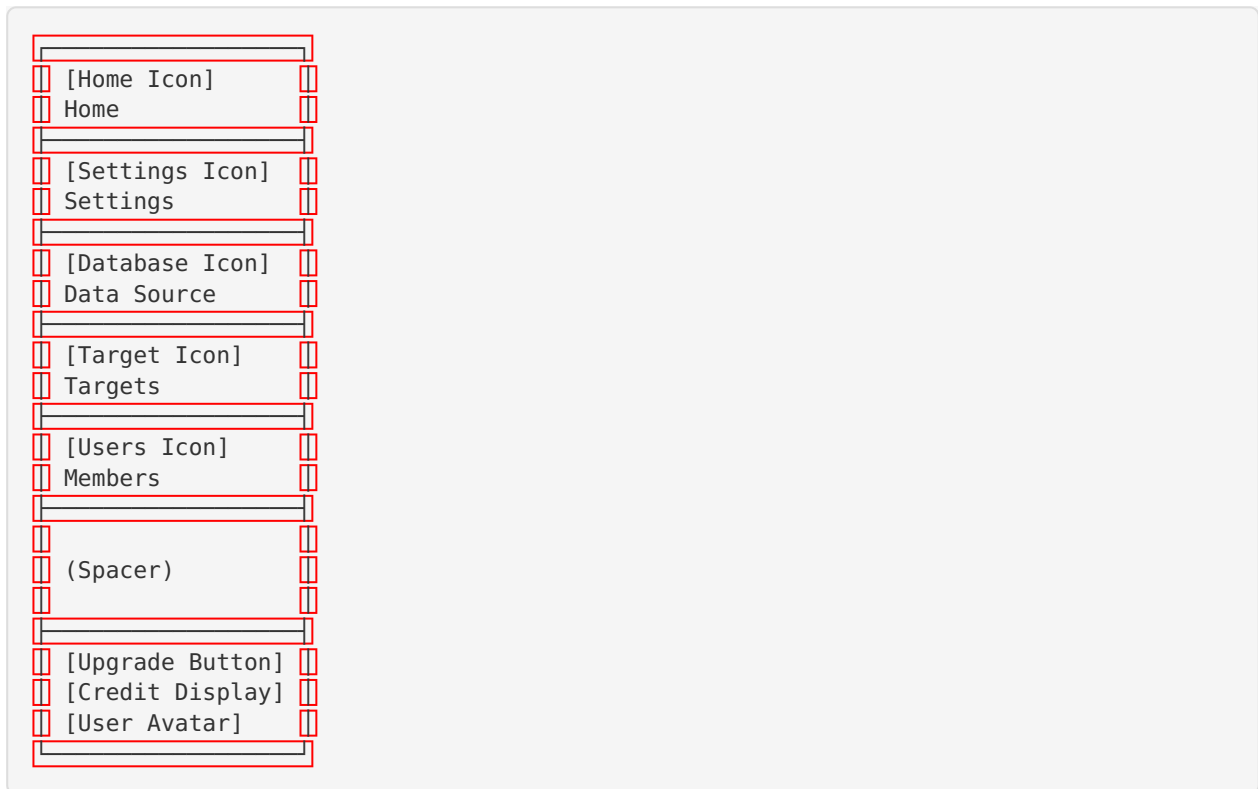
- **Logo:** Abacus.AI branding (left-aligned)
- **Navigation Items:**
 - Workspace
 - Tasks
 - Analytics
 - Tools
- **Right Section:**
 - User avatar/profile button
 - Settings icon
 - Notifications (optional)

Design Characteristics

- **Background:** White with subtle backdrop blur (`bg-white/60 backdrop-blur-xl`)
 - **Border:** Bottom border with light color (`border-b border-white/40`)
 - **Height:** Fixed height (~60-70px)
 - **Typography:** Clean sans-serif, medium weight for nav items
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3. Sidebar Component

Navigation Structure



Key Features

- **Width:**
 - Mobile: 60px (icon-only)
 - Desktop: 220px (icon + label)
- **Background:** White with backdrop blur (`bg-white/60 backdrop-blur-xl`)
- **Border:** Right border (`border-r border-white/40`)
- **Active State:** Highlighted background for current page
- **Bottom Section:**
 - Upgrade/billing button
 - Credit usage display
 - User avatar with dropdown

Navigation Items

1. **Home** - Dashboard/workspace overview
2. **Settings** - Account settings
3. **Data Source** - Data management
4. **Targets** - Goal/target configuration
5. **Members** - Team management

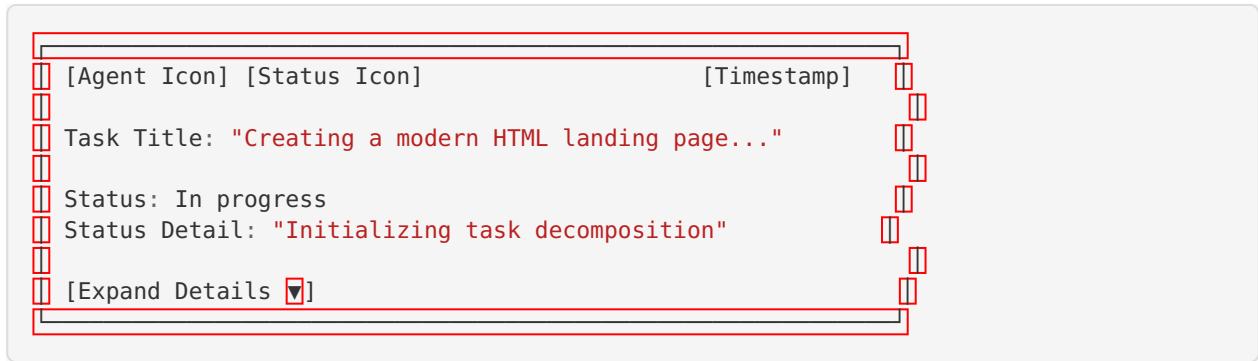
Design Characteristics

- **Icons:** 16px size, consistent style (Lucide icons)
- **Typography:** Small, clean sans-serif
- **Hover State:** Subtle background color change

- **Active State:** Distinct background color (purple/violet tint)

4. Main Content Area - Task Cards

Task Card Pattern (Critical Component)



Visual Hierarchy

- 1. Header Row:**
 - Agent icon (left)
 - Status icon (loading spinner / checkmark / error)
 - Timestamp (right)
- 2. Title:**
 - Large, bold text
 - Truncated with ellipsis if too long
 - Full text on hover
- 3. Status Section:**
 - Primary status: "In progress" / "Completed" / "Failed"
 - Secondary status detail: Current operation description
- 4. Expandable Details:**
 - Chevron icon to expand/collapse
 - Shows logs, artifacts, and tool usage when expanded

Status States & Color Coding

Status	Color	Icon	Description
Queued	Gray	Clock	Task waiting to start
Planning	Purple	Brain	Agent analyzing requirements
In Progress	Purple	Spinner	Active execution
Completed	Green	Checkmark	Successfully finished
Failed	Red	X	Error occurred
Cancelled	Orange	Stop	User cancelled

Expanded View

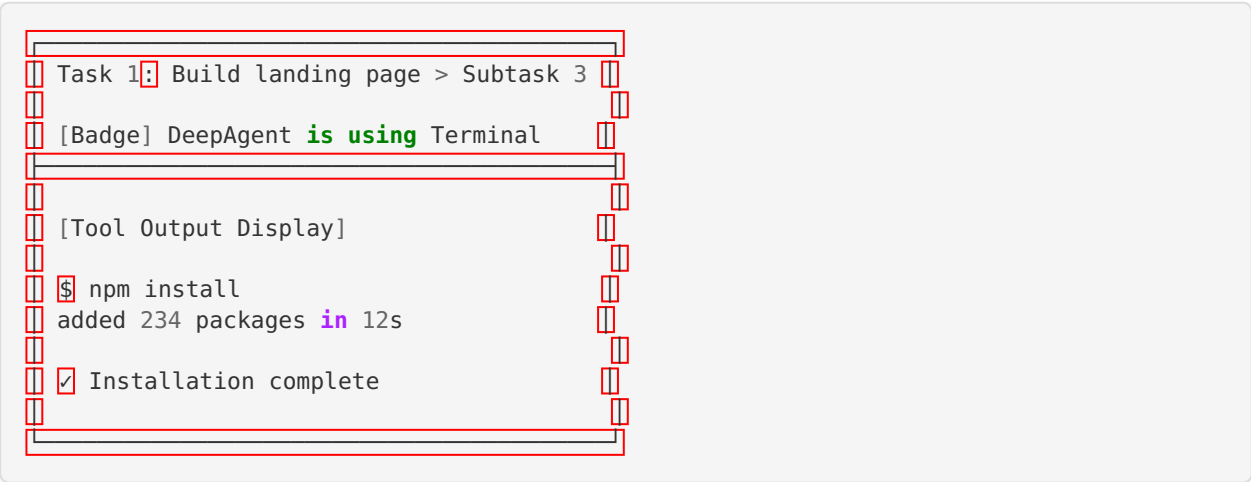
- When expanded, task card shows:
- **Subtask List:** Hierarchical view of all subtasks
 - **Logs:** Chronological event stream
 - **Artifacts:** Generated files with download buttons
 - **Tool Calls:** List of tools used with parameters
 - **Reasoning Traces:** Agent’s decision-making process

5. Tool Visibility Panel (Right Side)

Purpose

Shows real-time tool usage during task execution. Opens conditionally when agent is actively using a tool.

Structure



Header Format

- **Breadcrumb:** “Task X: [task name] > [current subtask]”

- **Badge:** “DeepAgent is using [Tool Name]”
- **Visual Separation:** Clear border/background distinction

Tool-Specific Output Display

1. Search Tool

- **Format:** Image grid with clickable sources
- **Elements:** Thumbnail images, source URLs, titles
- **Interaction:** Click to open source in new tab

2. Terminal

- **Format:** Monospace text with ANSI color coding preserved
- **Elements:** Command prompt, output, status indicators
- **Features:** Scrollable, copy-to-clipboard

3. Code Editor

- **Format:** Syntax-highlighted file content
- **Elements:** Line numbers, file path header, language indicator
- **Features:** Expandable, downloadable

4. File Operations

- **Format:** Git-style diff view
- **Elements:**
 - Green lines: Additions (+)
 - Red lines: Deletions (-)
 - Gray lines: Context
- **Features:** Side-by-side or unified diff view

Design Characteristics

- **Width:** ~400-500px on desktop
- **Background:** Light gray (bg-gray-50)
- **Border:** Left border to separate from main content
- **Scroll:** Independent scroll from main content
- **Animation:** Slide-in from right when opening

6. Progress Tracking System

Bottom Status Bar (Sticky)

Task 1, Subtask 3 - Installing dependencies...

Key Features

- **Position:** Fixed to bottom of viewport
- **Format:** “Task X, Subtask Y - [Current Operation]”
- **Updates:** Real-time as subtasks progress

- **Persistence:** Visible across all views
- **Background:** Semi-transparent with backdrop blur
- **Height:** ~40-50px

Progress Counter Behavior

- Increments in real-time as work progresses
- Shows hierarchical task/subtask relationship
- Displays current operation description
- Smooth transitions between states

7. File Operation Badges

Badge Format

```
[Action] ~/path/to/file [Status Icon]
```

Examples

- Written ✓ ~/app/page.tsx
- Updated ✓ ~/components/header.tsx
- Running ~/scripts/deploy.sh

Badge States

State	Color	Icon	Description
Queued	Gray outline	-	Waiting to execute
Running	Blue	Spinner	Currently executing
Completed	Green	Checkmark	Successfully finished
Failed	Red	X	Error occurred

Interactive Features

- **Expandable:** Click (:) icon to see full command details
- **Hover State:** Reveals download button for artifacts
- **Preview:** Click badge to open file preview modal
- **Copy Path:** Right-click to copy file path

Visual Design

- **Typography:** Monospace for file paths
- **Padding:** Compact, pill-shaped badges
- **Border Radius:** Rounded corners (8px)
- **Shadow:** Subtle drop shadow for depth

8. Status Indicators Throughout Execution

Phase 1: Planning

```
Status: "Planning"
Status Detail: "Understanding requirements"
Status Detail: "Weighing the possibilities"
Status Detail: "Creating execution plan"
```

Phase 2: Execution

```
Status: "In Progress"
- Real-time tool badges
- File operation badges
- Subtask progression counter
- Terminal output streaming
```

Phase 3: Completion

```
Status: "Completed"
- Green checkmark in task card
- Summary of all file operations
- Action CTAs (Preview, Deploy, Download)
- Credits used display
```

9. Color Scheme & Typography

Color Palette

Primary Colors

- **Purple/Violet:** #7C3AED - Actions, active states, primary buttons
- **Success Green:** #10B981 - Checkmarks, completions, success states
- **Error Red:** #EF4444 - Errors, cancellations, warnings
- **Warning Orange:** #F59E0B - Cautions, pending actions

Neutral Colors

- **Background Light:** #F3F4F6 - Page backgrounds
- **Background Medium:** #E5E7EB - Card backgrounds
- **Border:** #D1D5DB - Dividers, borders
- **Text Primary:** #111827 - Main text
- **Text Secondary:** #6B7280 - Supporting text

Accent Colors

- **Blue:** #3B82F6 - Links, info states
- **Teal:** #14B8A6 - Highlights, special states

Typography

Font Families

- **Sans-serif:** Inter, system-ui - UI elements, labels, body text

- **Monospace:** 'Fira Code', 'Courier New' - Code, file paths, terminal output
- **Serif:** (Optional) - Marketing content

Font Sizes

- **h1:** 2.5rem (40px) - Page titles
- **h2:** 2rem (32px) - Section headers
- **h3:** 1.5rem (24px) - Task card titles
- **h4:** 1.25rem (20px) - Subsection headers
- **Body:** 1rem (16px) - Main content
- **Small:** 0.875rem (14px) - Metadata, timestamps
- **Tiny:** 0.75rem (12px) - Badges, labels

Font Weights

- **Regular:** 400 - Body text
 - **Medium:** 500 - Navigation items, labels
 - **Semibold:** 600 - Headings, emphasis
 - **Bold:** 700 - Strong emphasis, CTAs
-

10. Interactive Elements

Expandable/Collapsible Components

- **Task Cards:** Click to expand/collapse details
- **Command Badges:** Click (:) to see full command
- **Subtask Lists:** Hierarchical expand/collapse
- **Log Sections:** Collapsible by category

Inline Code Editor

- **Syntax Highlighting:** Language-specific color coding
- **Line Numbers:** Left gutter with line numbers
- **Copy Button:** Top-right corner for quick copy
- **Download Button:** Save file locally

Preview Modal

- **Device Emulation:** Desktop, tablet, mobile views
- **Zoom Controls:** Zoom in/out
- **Refresh:** Reload preview
- **Open in New Tab:** Full-screen view

Copy-to-Clipboard

- **Code Blocks:** Copy button on hover
- **File Paths:** Right-click context menu
- **Terminal Output:** Select and copy
- **Feedback:** Toast notification on copy

Download Functionality

- **Artifacts:** Download button on hover

- **Logs:** Export full execution log
- **Reports:** Download summary report
- **Batch Download:** Download all artifacts as ZIP

Share/Edit Buttons

- **Share:** Generate shareable link
 - **Edit:** Open in code editor
 - **Fork:** Create copy for modification
 - **Feedback:** Thumbs up/down on messages
-

11. Real-time Update Animations

Smooth Transitions

- **Status Changes:** Fade transition (300ms)
- **Color Changes:** Smooth color interpolation
- **Size Changes:** Ease-in-out scaling

Fade-in for New Elements

- **New Subtasks:** Fade in from top (200ms)
- **New Messages:** Slide in from bottom (250ms)
- **New Badges:** Pop in with scale (150ms)

Pulse Effect

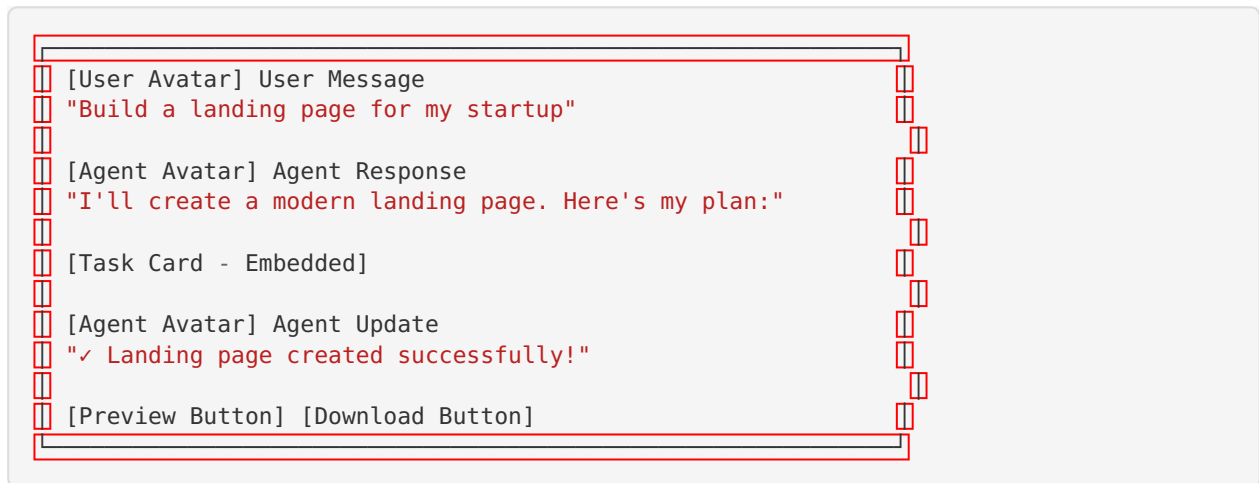
- **Active Operations:** Subtle pulse on active badges
- **Loading States:** Pulsing opacity (1s cycle)
- **Notifications:** Attention-grabbing pulse

Loading Animations

- **Spinner:** Rotating circle (1s rotation)
 - **Vertical Bars:** Brand-consistent loading indicator
 - **Skeleton Screens:** Shimmer effect for loading content
 - **Progress Bars:** Smooth fill animation
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12. Chat Interface

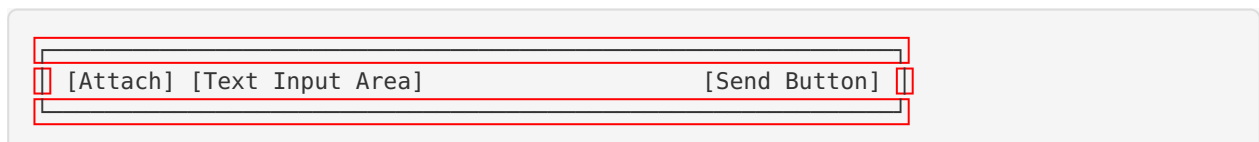
Structure



Message Types

1. **User Messages:**
 - Right-aligned (or left with avatar)
 - User avatar
 - Timestamp
 - Edit/delete options
2. **Agent Messages:**
 - Left-aligned
 - Agent avatar (robot icon)
 - Timestamp
 - Feedback buttons (thumbs up/down)
3. **System Messages:**
 - Centered
 - Gray background
 - Informational (e.g., "Task started", "Task completed")
4. **Embedded Task Cards:**
 - Full-width within chat
 - Interactive (expandable)
 - Real-time status updates

Input Area



- **Attach Button:** Upload files, images
- **Text Input:** Multi-line, auto-expanding
- **Send Button:** Disabled when empty, enabled when text present
- **Keyboard Shortcut:** Enter to send, Shift+Enter for new line

13. Dashboard/Workspace Overview

Layout

Multi-LLM Workspace

Orchestrate **complex** tasks across multiple AI providers

[Compose Task] [Live Monitor] [Task History] [Analytics]

New Task Specification

Task Title *

[Input: e.g., Build a React todo app...]

Detailed Description

[Textarea: Additional context...]

LLM Configuration

Primary LLM Provider: [Auto-Route (Recommended) ▼]

Task Type: [General Action ▼]

Priority Level: [Medium Priority ▼]

Execution Options

☒ Task Decomposition

[Submit Task]

Provider Status

[Abacus AI - Online] [ChatGPT - Online] [Claude - Online]

Today's Usage

API Calls: 2,847 | Cost: \$12.34 | Avg Time: 1.2s

Key Sections

1. Header:

- Workspace name
- Description
- Stats (completed tasks, active tasks, total cost)

2. Tab Navigation:

- Compose Task
- Live Monitor
- Task History
- Analytics

3. Task Composition Form:

- Task title (required)
- Detailed description (optional)
- LLM configuration (auto-route recommended)
- Task type dropdown

- Priority level dropdown
- Execution options (task decomposition toggle)
- Submit button

4. **Provider Status:**

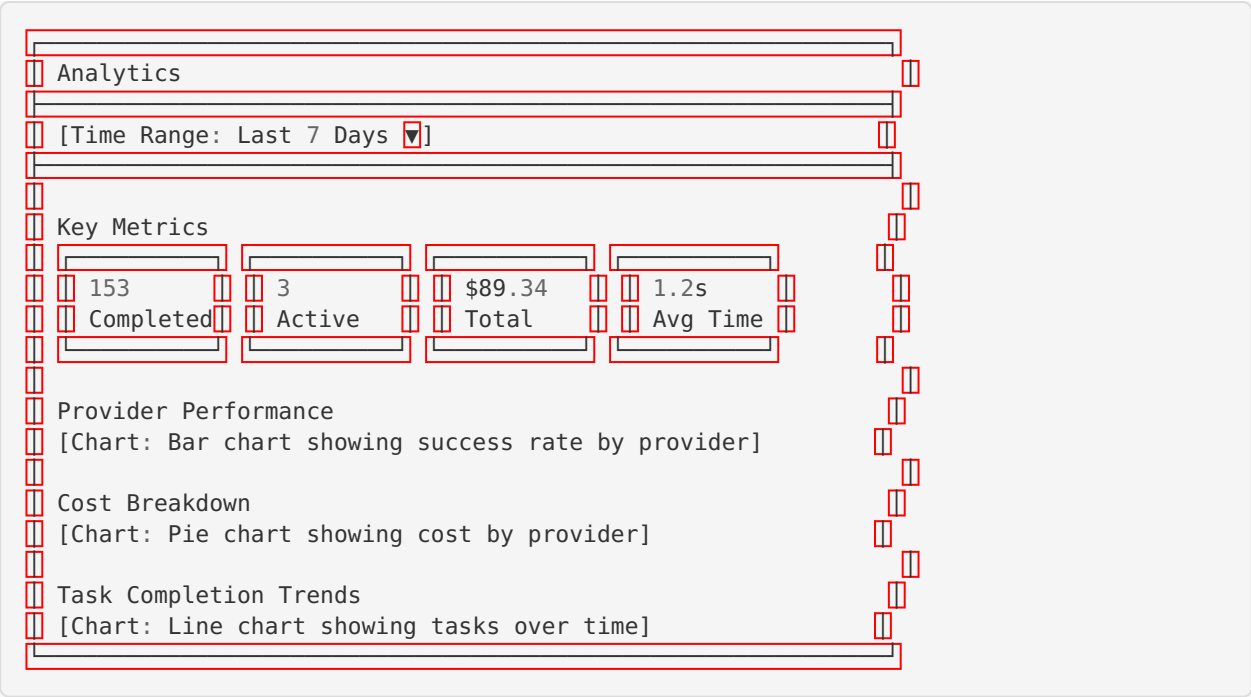
- Real-time status badges for each LLM provider
- Color-coded: Green (online), Red (offline), Yellow (slow)

5. **Usage Stats:**

- Today’s API calls
- Cost estimate
- Average response time

14. Analytics View

Metrics Dashboard



Charts & Visualizations

- **Bar Charts:** Provider performance comparison
- **Pie Charts:** Cost distribution
- **Line Charts:** Trends over time
- **Heatmaps:** Usage patterns by time of day

15. Settings Page

Account Settings Structure

Settings

Account Settings
Manage your account preferences

[Settings options will be available here.]

Expected Settings Sections

- 1. **Profile:**
 - Name, email, avatar
 - Password change
 - Two-factor authentication
- 2. **Billing:**
 - Current plan
 - Usage credits
 - Payment method
 - Billing history
- 3. **API Keys:**
 - LLM provider API keys
 - Integration tokens
 - Webhooks
- 4. **Preferences:**
 - Theme (light/dark)
 - Notifications
 - Language
 - Timezone
- 5. **Team:**
 - Members list
 - Roles & permissions
 - Invitations

16. Navigation Patterns

Primary Navigation

- **Sidebar:** Main navigation hub
- **Header:** Secondary navigation and user actions
- **Breadcrumbs:** Hierarchical navigation within sections
- **Tabs:** Section-specific navigation (e.g., Compose Task, Task History)

Navigation Behavior

- **Active State:** Highlighted in sidebar
- **Hover State:** Subtle background change
- **Click:** Instant navigation (no page reload)
- **Back Button:** Browser back button supported

Deep Linking

- **Task URLs:** `/dashboard/tasks/{taskId}`
 - **Settings URLs:** `/dashboard/settings/{section}`
 - **Analytics URLs:** `/dashboard/analytics?range=7d`
-

17. Responsive Design Patterns

Desktop (>1024px)

- Full sidebar with labels
- Tool visibility panel on right
- Multi-column layouts
- Expanded task cards

Tablet (768px - 1024px)

- Collapsible sidebar (icon-only by default)
- Tool panel overlays main content
- Two-column layouts
- Compact task cards

Mobile (<768px)

- Bottom navigation bar
 - Full-screen tool panel
 - Single-column layouts
 - Stacked task cards
 - Hamburger menu for sidebar
-

18. Accessibility Features

Keyboard Navigation

- **Tab:** Navigate through interactive elements
- **Enter/Space:** Activate buttons and links
- **Escape:** Close modals and panels
- **Arrow Keys:** Navigate lists and menus

Screen Reader Support

- **ARIA Labels:** Descriptive labels for all interactive elements
- **ARIA Live Regions:** Announce real-time updates

- **Semantic HTML:** Proper heading hierarchy, landmarks
- **Alt Text:** Descriptive text for images and icons

Visual Accessibility

- **High Contrast Mode:** Support for high contrast themes
 - **Focus Indicators:** Clear focus outlines
 - **Color Blindness:** Color is not the only indicator of state
 - **Font Scaling:** Supports browser font size adjustments
-

19. Performance Optimizations

Real-time Updates

- **WebSocket Connection:** Low-latency streaming
- **Debouncing:** Prevent excessive updates
- **Throttling:** Limit update frequency
- **Batching:** Group multiple updates

Rendering Optimizations

- **Virtual Scrolling:** For long lists
- **Lazy Loading:** Load content as needed
- **Code Splitting:** Load routes on demand
- **Image Optimization:** Compressed, responsive images

Caching Strategies

- **API Response Caching:** Reduce redundant requests
 - **Static Asset Caching:** Cache CSS, JS, images
 - **Service Worker:** Offline support
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20. Key Takeaways for Mindous Redesign

Must Implement (Critical)

1.  **Task Card Pattern** - Visual task status with clear states
2.  **Tool Visibility Panel** - Right-side panel for real-time tool usage
3.  **Progress Counter** - Sticky bottom bar "Task X, Subtask Y"
4.  **File Operation Badges** - Visual tracking of file changes
5.  **Real-time Status Updates** - Live updates without refresh
6.  **Color Scheme** - Purple primary, green success, red error
7.  **Sidebar Navigation** - Left sidebar with icon + label

Should Implement (High Priority)

1.  **Diff Viewer** - Git-style file comparisons
2.  **Code Editor Integration** - Syntax-highlighted viewing
3.  **Artifact Management** - Download buttons for files
4.  **Pause/Cancel Controls** - User control over execution

5.  **Feedback Buttons** - Thumbs up/down on tasks

Nice to Have (Phase 2)

1.  **Search within Logs** - Filter/search log entries
2.  **Export Execution Report** - Download full trace
3.  **Video Recording** - Replay execution visually

21. Design System Summary

Component Library

- **Buttons:** Primary, secondary, ghost, danger
- **Cards:** Task cards, info cards, stat cards
- **Badges:** Status badges, file operation badges, provider badges
- **Inputs:** Text, textarea, select, checkbox, toggle
- **Modals:** Confirmation, preview, settings
- **Panels:** Sidebar, tool visibility, settings
- **Lists:** Task lists, log lists, file lists
- **Charts:** Bar, line, pie, heatmap

Spacing System

- **xs:** 4px
- **sm:** 8px
- **md:** 16px
- **lg:** 24px
- **xl:** 32px
- **2xl:** 48px

Border Radius

- **sm:** 4px
- **md:** 8px
- **lg:** 12px
- **xl:** 16px
- **full:** 9999px (pills)

Shadows

- **sm:** 0 1px 2px rgba(0,0,0,0.05)
- **md:** 0 4px 6px rgba(0,0,0,0.1)
- **lg:** 0 10px 15px rgba(0,0,0,0.1)
- **xl:** 0 20px 25px rgba(0,0,0,0.1)

Conclusion

The Abacus.AI platform demonstrates a mature, transparency-focused design system that prioritizes user trust through comprehensive visibility into agent operations. The key differentiator is the **real-**

time progress tracking and **tool usage visibility**, which keeps users informed at every step of task execution.

For the Mindous redesign, the priority should be implementing the **Task Card Pattern**, **Tool Visibility Panel**, and **Progress Counter** as these are the foundational elements that define the user experience. The color scheme (purple primary, green success) and typography (Inter + monospace) should be adopted to maintain visual consistency with the reference platform.

Next Steps:

1. Audit current Mindous build against this mapping
2. Identify components to remove (agent orchestration-specific)
3. Identify components to keep/modify (core UI patterns)
4. Create implementation roadmap for redesign

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